SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4 **Duration: 1 Hour**
Total: Marks:40

Annexure A

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Code of Conduct:

I N. Chawran Aryan certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

N. Chawran Aryan
Signature of the student:

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns a general rule or model from a set of labelled examples and uses it to predict outcomes for unseen patients.

For this healthcare dataset, a **supervised classification approach** is suitable because the target variable is whether the patient has diabetes.

Target variable:

- Diabetes → Diabetic / Non-Diabetic

Important features:

1. **Glucose Level** – A high glucose level is one of the strongest indicators of diabetes.
2. **BMI** – Higher BMI is associated with increased risk of developing type-2 diabetes.
3. **Age** – Diabetes risk generally increases with age.
4. **Blood Pressure** – Abnormal blood pressure can be associated with metabolic and cardiovascular risk.
5. **Cholesterol** – Abnormal cholesterol levels can provide additional information about metabolic health.
6. **Family History** – A family history of diabetes increases an individual's risk.

Thus, a classification model such as **Decision Tree, Logistic Regression, or Naïve Bayes** can learn the relationship between these features and the diabetes target.

(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients, resulting in **class imbalance**.

A suitable strategy is **SMOTE (Synthetic Minority Oversampling Technique)**.

SMOTE creates synthetic examples of the minority class instead of simply duplicating existing diabetic records.

Advantages:

- Increases representation of diabetic cases.
- Helps the classifier learn minority-class patterns.
- Reduces bias toward the majority class.
- Can improve **Recall and F1-score** for diabetic patients.

The SMOTE operation should be performed **only on the training set**, not before the train-test split. This prevents data leakage.

For a medical application, **class weights** can also be used as an alternative or complementary approach, particularly when synthetic samples may not be clinically appropriate.

(c) 70:30 Split and Preprocessing

The dataset can be divided as:

- **70% → Training data**
- **30% → Testing data**

The **70:30 split** provides enough data to train the model while retaining a sufficiently large independent test set for evaluating its performance.

For a medical application, the test set should represent unseen patients.

Preprocessing steps:

1. **Handle missing values**
   - Numerical missing values → median imputation.
   - Categorical missing values → mode imputation.
2. **Encoding**
   - Convert categorical features such as family history into numerical form.
   - Example:
     - Yes → 1
     - No → 0

3. **Normalization/Standardization**
    - Features such as glucose, blood pressure, cholesterol and BMI can have different scales.
    - Standardization transforms them to a comparable scale.
4. **Class balancing**
    - Apply SMOTE only to the training data.
5. **Train-test split**
    - Use stratification so that diabetic/non-diabetic proportions remain similar in both sets.

Task 2: Decision Tree for Diagnosis
(a) Decision Tree Classifier
A **Decision Tree classifier** can be used to predict whether a patient is diabetic.
A possible tree structure is:

```
                Glucose Level
               /           \
           Low             High
          /                   \
      BMI                      BMI
     /   \                   /   \
  Normal  High             Normal  High
    |      |               |      |
 Non-Diabetic Risk          Risk   Diabetic
```

The exact tree depends on the actual dataset.
Root Node Selection
The root node is selected using a measure such as **Information Gain** or **Gini Index**.
Information Gain:

$$
IG(S,A)=H(S)-\sum_{v\in Values(A)}
\frac{|S_v|}{|S|}H(S_v)
$$

where:

- ($H(S)$) = entropy of the original dataset
- ($H(S_v)$) = entropy after splitting using attribute A

The feature with the **highest Information Gain** is selected as the root.
Alternatively, using Gini Index:

$$
Gini(S)=1-\sum_{i=1}^{n}p_i^2
$$

The feature producing the **lowest weighted Gini impurity** is preferred.
For this medical problem, **Glucose Level would commonly be expected to be a strong root candidate**, but the actual root must be determined from the dataset's calculated Information Gain/Gini values.

(b) Model Evaluation
The Decision Tree should be evaluated using:
Accuracy

$$
Accuracy=\frac{TP+TN}{TP+TN+FP+FN}
$$

]
Precision
[
Precision=\frac{TP}{TP+FP}
]
Recall
[
Recall=\frac{TP}{TP+FN}
]
F1-Score
[
F1=2\times\frac{Precision\times Recall}
{Precision+Recall}
]
Where:

- **TP** = correctly identified diabetic patients
- **TN** = correctly identified non-diabetic patients
- **FP** = non-diabetic patients incorrectly classified as diabetic
- **FN** = diabetic patients incorrectly classified as non-diabetic

False Negatives
A **False Negative** occurs when:
Actual patient = Diabetic
Predicted = Non-Diabetic
False negatives are particularly dangerous in this application because a diabetic patient may remain undiagnosed and untreated.
How to reduce False Negatives
The model should prioritize **Recall/Sensitivity** for the diabetic class.
Possible methods:

- Use class weights.
- Apply appropriate oversampling such as SMOTE.
- Adjust the classification threshold where applicable.
- Tune Decision Tree parameters.
- Use recall/F1 rather than accuracy alone for model selection.

Clinically, it is generally preferable to investigate a possible diabetes case further than to incorrectly classify a diabetic patient as healthy.

(c) Post-Pruning
A fully grown Decision Tree may overfit the training data.
Cost-complexity pruning controls this by minimizing:
[
R_\alpha(T)=R(T)+\alpha|T|
]
where:

- (R(T)) = tree error
- (|T|) = number of terminal nodes
- (\alpha) = complexity parameter

Comparison

Model	Advantage	Disadvantage
Pre-pruned Tree	Faster and simpler	May stop too early
Post-pruned Tree	Removes unnecessary branches after training	Requires additional tuning
Fully grown Tree	Captures complex patterns	Higher risk of overfitting

For clinical deployment, the **pruned Decision Tree** is generally preferable if it provides comparable or better test performance because it is simpler, easier to interpret and less prone to overfitting.

Task 3: Statistical Learning for Risk Stratification
(a) Logistic Regression
Logistic Regression can be applied as a statistical learning model for predicting diabetes.
The probability of diabetes is:

$$
P(Y=1)=\frac{1}{1+e^{-z}}
$$

where

$$
z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n
$$

For example:

$$
z=\beta_0+\beta_1(Glucose)+\beta_2(BMI)+
\beta_3(Age)+\beta_4(BP)+\beta_5(Cholesterol)
$$

The output is a probability between 0 and 1.
Example:
Probability = 0.82
If the chosen threshold is 0.5:
0.82 > 0.5 → Diabetic
Comparison with Decision Tree

Metric	Decision Tree	Logistic Regression
Accuracy	Calculate from test set	Calculate from test set
AUC-ROC	Calculate from probabilities	Calculate from probabilities
Interpretability	Very high	High
Non-linear relationships	Handles naturally	Requires feature engineering
Probability output	Possible	Directly provided
Clinical usefulness	Easy rule-based explanation	Easy statistical interpretation

Important: Since the question does not provide the actual patient dataset, numerical Accuracy and AUC-ROC values cannot be truthfully reported here. They must be calculated from the supplied dataset. The question explicitly asks for those performance comparisons.
When is a statistical model preferable?
Logistic Regression is preferable when:

- The relationship between predictors and outcome is approximately linear in the log-odds.
- Probability estimates are important.
- A simpler mathematical model is desired.
- Interpretability is important.
- The dataset is relatively small.

- Coefficient-based explanation is useful.

(b) Feature Importance
The top three predictors should be obtained separately from both models.
Decision Tree
Decision Tree feature importance is usually based on the total reduction in impurity contributed by each feature.
Expected important predictors may include:

1. Glucose Level
2. BMI
3. Age

However, the **actual top three must be calculated from the trained tree**.
Logistic Regression
For Logistic Regression, feature importance can be examined using standardized coefficients:

$$Importance_i=|\beta_i|$$

The three features with the largest absolute standardized coefficients can be considered the strongest predictors.
Do they agree?
They may or may not agree.
If both models identify **glucose, BMI and age** as important, it suggests that these variables have a strong and consistent relationship with diabetes.
If they differ, it can occur because:

- Decision Trees capture non-linear relationships.
- Logistic Regression assumes a linear relationship in the log-odds.
- Correlated features may distribute importance differently.
- Tree-based importance can be affected by feature structure.

Therefore, agreement between models strengthens confidence, while disagreement indicates that further statistical and clinical investigation is required.

Task 4: Reinforcement Learning for Treatment Recommendation
The assessment asks to formulate treatment recommendation as an **MDP**, apply Q-Learning for at least five episodes, show Q-table updates and extract the final policy.
(a) MDP Model
A **Markov Decision Process (MDP)** can be represented as:

$$MDP=(S,A,P,R,\gamma)$$

States
States represent the patient's health condition.
Example:
S0 = Normal / Low Risk
S1 = Prediabetic / Moderate Risk
S2 = Diabetic / High Risk
S3 = Improving
S4 = Stable
Actions

The given actions are:
A1 = Diet
A2 = Exercise
A3 = Medication
A4 = Monitor

Reward Function

The reward should represent improvement in patient health.
Example:

Patient outcome	**Reward**
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-5
Unsafe treatment outcome	-10

The reward encourages the agent to select actions that improve patient health while avoiding harmful decisions.

Transition Dynamics

The transition probability describes how a patient moves from one health state to another after an action.
Example:
Prediabetic + Diet → Improving
Prediabetic + Exercise → Improving
Diabetic + Medication → Stable
Stable + Monitor → Stable
In a real clinical system, these transitions must be learned from validated clinical data and should not be assumed arbitrarily.

(b) Q-Learning

The Q-Learning update equation is:

$$Q(s,a)\leftarrow Q(s,a)+\alpha[r+\gamma\max_{a'}Q(s',a')-Q(s,a)]$$

Assume:

- Learning rate ($\alpha=0.5$)
- Discount factor ($\gamma=0.9$)
- Initial Q-values = 0

Iteration 1

Suppose:
State = Prediabetic
Action = Diet
Reward = +5
Next State = Improving
Assume the maximum Q-value of the next state is 0.

$$Q(S1,Diet)=0+0.5[5+0.9(0)-0]$$

[

$Q(S1,Diet)=2.5$

]

So:

$Q(S1,Diet) = 2.5$

Iteration 2

Suppose:

State = Prediabetic

Action = Exercise

Reward = +5

Next State = Improving

Again, maximum next-state Q-value = 0.

[

$Q(S1,Exercise)=0+0.5[5+0.9(0)-0]$

]

[

$Q(S1,Exercise)=2.5$

]

So:

$Q(S1,Exercise) = 2.5$

Third State-Action Pair

Suppose:

State = Diabetic

Action = Medication

Reward = +10

Next State = Stable

Assume maximum Q-value of Stable = 0.

[

$Q(S2,Medication)$

$=0+0.5[10+0.9(0)-0]$

]

[

$Q(S2,Medication)=5$

]

So:

$Q(S2,Medication) = 5$

Example Q-table

State	Diet	Exercise	Medication	Monitor
Normal	0	0	0	0
Prediabetic	2.5	2.5	0	0
Diabetic	0	0	5.0	0
Improving	0	0	0	0
Stable	0	0	0	0

The agent should be trained for **at least five patient episodes**, as required by the assessment. The actual final Q-table depends on the transition and reward data used.

Convergence Criterion

Training can be considered converged when:

[

$\max |Q_{new}(s,a)-Q_{old}(s,a)| < \epsilon$

]
for example:
[
\epsilon=0.001
]
for several consecutive episodes.

(c) Final Learned Policy
The policy is obtained by selecting the action with the highest Q-value:
[
\pi(s)=\arg\max_a Q(s,a)
]
Example:

State	**Highest Q-value**	**Recommended action**
Normal	Monitor	Monitor
Prediabetic	2.5	Diet/Exercise
Diabetic	5.0	Medication*
Improving	Highest learned value	Continue appropriate intervention
Stable	Highest learned value	Monitor

*In a real healthcare deployment, the RL agent **must not independently prescribe medication**. Medication decisions require qualified clinical oversight.

Ethical Considerations
1. Patient Safety
A reinforcement-learning system can make incorrect decisions if trained on inadequate or biased data. Therefore, it should operate with **clinician supervision and safety constraints**.
2. Bias
If the training data under-represents certain demographic or clinical groups, the agent may produce unfair recommendations.
Regular bias and subgroup-performance testing are required.
3. Explainability
Healthcare professionals need to understand why an action was recommended.
The system should provide:

- Patient state
- Relevant risk factors
- Selected action
- Expected outcome
- Reason for recommendation

4. Human-in-the-loop
The AI should be used as a **decision-support system**, not as an autonomous replacement for doctors.
5. Privacy
Patient medical information must be securely stored and processed with appropriate privacy and access controls.
6. Validation
Before deployment, the model should undergo extensive validation and clinical evaluation.